

Ing. **MICHAL NEORAL** Ph.D.



PROFILE

Computer Vision Engineer and Researcher with expertise in dense motion estimation, optical flow, and long-term pixel tracking. Skilled in deep learning and in developing efficient, scalable solutions for complex visual perception tasks. Experienced in industrial R&D collaborations, including projects with Toyota.

CONTACT DETAILS

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PERSONAL INFORMATION

Citizenship: **Czech Republic**
Family: **Single without children**
Languages: **English, Czech**

SKILLS

- Machine Learning, Tracking
- Optical Flow, Motion Segmentation
- Image Processing, Object Detection, Data Visualization
- Communication and team collaboration

SOFTWARE SKILLS

- Python, PyTorch, OpenCV, SciPy, NumPy, JAX
- Linux, Bash, ~~TeX~~ LaTeX, Git
- MS Office and its alternatives

Previous experience with

- CUDA, TensorFlow
- Matlab, R, SQL, LabView
- C, C++, C#, Java

EXPERIENCE

RESEARCH DEVELOPER at *Czech Technical University in Prague.*

project **FUNDIVE**

2026.05–pres.

◇ Developing robust open-set and closed-set recognition methods for fungal species, where classification is challenging due to fine-grained visual differences, high natural variability, imbalanced data, and unknown species at inference time.

RESEARCH DEVELOPER at *Czech Technical University in Prague.*

project **LOCAL CLS** (in parallel with the above)

2026.05–pres.

◇ Designed and implemented a multi-stage knowledge-distillation method for fine-grained image retrieval, including a novel learned-threshold attention pooling mechanism for compact descriptors from frozen vision-language models.

RESEARCH DEVELOPER at *Czech Technical University in Prague.*

project **CEDMO**

2026.02–2026.05

◇ Built a user-facing tool for automated analysis of billboard video recordings, associating detected people and textual content across many captured billboard instances to support large-scale media monitoring.

RESEARCH DEVELOPER at *Czech Technical University in Prague.*

project **AISEE**

2025.02–2026.01

◇ Developing an AI surveillance system for large-scale video object detection and classification with fast, incremental learning of user-defined classes.

RESEARCH DEVELOPER at *Czech Technical University in Prague.*

project **MEDIAN** (in parallel with the above)

2025.02–2026.01

◇ Building a system for automatic recognition of template videos from cam-glasses recordings.

RESEARCH SCIENTIST at *Czech Technical University in Prague.*

Comp. Vision research for **TOYOTA MOTOR EUROPE** **2017.07–2025.01**

◇ Developed a CNN-based optical flow method that jointly estimates occlusions and integrates temporal context, achieving state-of-the-art accuracy.
◇ Designed a deep model combining semantic and geometric cues to segment independently moving objects in videos captured from a moving platform.
◇ Introduced a fast, dense, long-term tracker that chains optical flows with reliability estimation, outperforming prior methods on multiple benchmarks.

UNIVERSITY TEACHER at *Czech Technical University in Prague.*

multiple courses (in parallel with all above)

2018.03–pres.

◇ Lab teacher for courses: Pattern Recognition and Machine Learning (RPZ), Digital Photography Processing (DIF), and Computer Vision Methods (MPV).
◇ In addition to teaching, developed a student homework evaluation tool for the RPZ course in Python.

INTERN at *Toyota Motor Europe* (Belgium).

2017.02–2017.06

◇ Development and implementation of monocular depth estimation method.

EXPERIENCE (JUNIOR AND PART-TIME)

RESEARCHER AND DEVELOPER at *Czech Technical University in Prague.*

Comp. Vision research for **TOYOTA MOTOR EUROPE** **2015.10–2017.02**

◇ Extension of the scene flow method to work with temporal consistency.

project **ELECTROLUX**

2015.06–2015.09

◇ Collab. on identification system of objects inserted into home appliances.

project **SYROTEK**

2014–2015

◇ Collab. on a robotic system for remote teaching and remote control.

project **CLOPEMA**

2012–2014

◇ Features extraction and automatic data collection from a moving garment.

EDUCATION

DOCTORAL DEGREE – Ph.D. (Philosophiæ Doctor)

2018–2025

Faculty of Electrical Engineering. *Czech Technical University in Prague*

◊ Informatics / Artificial Intelligence and Biocybernetics

Thesis: *Dense Motion Estimation in a Monocular Video*. ◊ supervisors prof. Ing. Jiří Matas, Ph.D. and Mgr. Jan Šochman, Ph.D.

MASTER IN COMPUTER VISION – Ing. (MSc equivalent)

2014–2017

Faculty of Electrical Engineering. *Czech Technical University in Prague*

◊ Open Informatics, specialization: Computer Vision and Image Processing

Thesis: *Object Scene Flow in Video Sequences*. ◊ supervisor Mgr. Jan Šochman, Ph.D.

◊ awarded with the Dean's Prize for outstanding master thesis.

BACHELOR IN ROBOTICS – Bc. (BSc equivalent)

2011–2014

Faculty of Electrical Engineering. *Czech Technical University in Prague*

◊ Cybernetics and Robotics, specialization: Robotics

Thesis: *Extraction of Features from Moving Garment*. ◊ supervisor Ing. Pavel Krsek, Ph.D.

PUBLICATIONS

Jonáš Šerých, [Michal Neoral](#), and Jiří Matas. (2025). [MFTIQ: MULTI-FLOW TRACKER WITH INDEPENDENT MATCHING QUALITY ESTIMATION](#). In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, February 2025, Tucson, Arizona USA.

[Michal Neoral](#), Jonáš Šerých, and Jiří Matas. (2024). [MFT: LONG-TERM TRACKING OF EVERY PIXEL](#). In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 6837–6847, January 2024, Waikoloa, Hawaii USA.

[Michal Neoral](#), Jan Šochman, and Jiří Matas. (2021). [MONOCULAR ARBITRARY MOVING OBJECT DISCOVERY AND SEGMENTATION](#). In *The 32nd British Machine Vision Conference*, November 2021, Online.

[Michal Neoral](#), Jan Šochman, and Jiří Matas. (2019). [CONTINUAL OCCLUSION AND OPTICAL FLOW ESTIMATION](#). Computer Vision – ACCV 2018. Ed. by C.V. Jawahar et al. Cham: Springer International Publishing, 2019, pp. 159–174. ISBN: 978-3-030-20870-7.

[Michal Neoral](#) and Jan Šochman. (2017). [OBJECT SCENE FLOW WITH TEMPORAL CONSISTENCY](#). In *22nd Computer Vision Winter Workshop (CVWW)*. Pattern Recognition and Image Processing Group; TU Wien & PRIP Club; Vienna; Austria, February 2017, ISBN: 978-3-200-04969-7.

PATENTS

Nikolay Chumerin, [Michal Neoral](#), Jan Šochman, and Jiří Matas. (2023). [MONOCULAR-VISION-BASED DETECTION OF MOVING OBJECTS](#). EP Patent EP4174770B1, filed Oct 28, 2021, and issued Nov 29, 2023. [GRANTED PATENT](#)

Nikolay Chumerin, Jonáš Šerých, [Michal Neoral](#), and Jiří Matas. (2024). [A METHOD AND SYSTEM FOR TRACKING A PIXEL IN A VIDEO](#). EP Patent EP4465239A1, filed May 16, 2023, and issued Nov 20, 2024. [PATENT APPLICATION](#)

Nikolay Chumerin, [Michal Neoral](#), Jan Šochman, and Jiří Matas. (2023). [MONOCULAR-VISION-BASED DETECTION OF MOVING OBJECTS](#). EP Patent EP4174770A1, filed Oct 28, 2021, and issued May 03, 2023. [PATENT APPLICATION](#)

Nikolay Chumerin, [Michal Neoral](#), Jan Šochman, and Jiří Matas. (2020). [METHODS FOR OPTICAL FLOW ESTIMATION](#). WO Patent WO2020088766A1, filed Oct 31, 2018, and issued May 07, 2020. [PATENT APPLICATION](#)

OTHER PUBLICATIONS

Matej Kristan, Jiří Matas, Pavel Tokmakov, Alan Lukežič, Michael Felsberg, Luka Čehovin Zajc, Khanh-Tung Tran, Xuan-Son Vu, Johanna Björklund, [Michal Neoral](#), Hyung Jin Chang, and others (2025). [THE THIRD VISUAL OBJECT TRACKING SEGMENTATION VOTS2025 CHALLENGE RESULTS](#). Proceedings of the IEEE/CVF International Conference on Computer Vision.

Adam Schmidt, Mert Asim Karaoglu, Soham Sinha, Mingang Jang, Ho-Gun Ha, Kyungmin Jung, Kyeongmo Gu, Ihsan Ullah, Hyunki Lee, Jonáš Šerých, [Michal Neoral](#), Jiří Matas, and others (2025). [POINT TRACKING IN SURGERY–THE 2024 SURGICAL TATTOOS IN INFRARED \(STIR\) CHALLENGE](#). arXiv preprint arXiv:2503.24306.